# EE3110 - Probability Foundations for Electrical Engineers
## Tutorial - 4

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1. Let $X$ and $Y$ be independent geometric random variables with $X \sim \text{Geo}\,(1/2)$ and $Y \sim \text{Geo}\,(1/3)$. Evaluate the following:

   (a) $\mathbb{P}(X > Y)$

   (b) $\mathbb{P}(X = Y)$

**Solution:** Clearly, $\mathbb{P}(X = x) = (1/2)^x \ \forall x \in \mathbb{N}$ and $\mathbb{P}(Y = y) = \left(\frac{1}{2}\right)(2/3)^y \ \forall y \in \mathbb{N}$

(a) $\mathbb{P}(X > Y) = \mathbb{P}(Y = 1)\mathbb{P}(X \geq 2) + \mathbb{P}(Y = 2)\mathbb{P}(X \geq 3) + \mathbb{P}(Y = 3)\mathbb{P}(X \geq 4) + \ldots$ Also, $\mathbb{P}(X \geq k) = \sum_{i=k}^{\infty}(1/2)^k = (1/2)^{k-1} \ \forall k \in \mathbb{N}$

$$\begin{aligned}
\text{Hence, } \mathbb{P}(X > Y) &= \sum_{i=1}^{\infty} \mathbb{P}(Y = i)\mathbb{P}(X \geq (i+1)) \\
&= \sum_{i=1}^{\infty} \left(\frac{1}{2}\right)\left(\frac{2}{3}\right)^i \times \left(\frac{1}{2}\right)^i \\
&= \left(\frac{1}{2}\right)\sum_{i=1}^{\infty}\left(\frac{1}{3}\right)^i \\
&= \boxed{\frac{1}{4} = 0.25}
\end{aligned}$$

(b)

$$\begin{aligned}
\mathbb{P}(X = Y) &= \mathbb{P}(Y = 1)\mathbb{P}(X = 1) + \mathbb{P}(Y = 2)\mathbb{P}(X = 2) + \mathbb{P}(Y = 3)\mathbb{P}(X = 3) + \ldots \\
&= \sum_{i=1}^{\infty} \mathbb{P}(Y = i)\mathbb{P}(X = i) \\
&= \sum_{i=1}^{\infty} \left(\frac{1}{2}\right)\left(\frac{2}{3}\right)^i \times \left(\frac{1}{2}\right)^i \\
&= \left(\frac{1}{2}\right)\sum_{i=1}^{\infty}\left(\frac{1}{3}\right)^i \\
&= \boxed{\frac{1}{4} = 0.25}
\end{aligned}$$

2. Let $X$ be a Poisson random variable with mean $\lambda > 0$. Given $X = x$, the random variable $Y$ is Binomial with parameters $(x, p)$. Define $Z = X - Y$. (a) What are the (marginal) distributions of $Y$ and $Z$? (b) Are $Y$ and $Z$ independent?

**Solution:** Given,

$$p_X(x) = \frac{e^{-\lambda} \cdot \lambda^x}{x!}\mathbb{1}_{\{x \geq 0\}}$$

$$p_{Y|X}(y|x) = \binom{x}{y} p^y (1-p)^{x-y} \cdot \mathbb{1}_{\{0 \le y \le x\}}$$

The joint p.m.f of Y and Z can be calculated as follows,

$$\begin{aligned}
p_{Y,Z}(y,z) &= p_{X,Y}(y+z, y) \\
&= p_X(y+z) \cdot p_{Y|X}(y|y+z) \quad \text{(By conditional p.m.f)} \\
&= \frac{e^{-\lambda} \cdot \lambda^{(y+z)}}{(y+z)!} \mathbb{1}_{\{y+z \ge 0\}} \cdot \binom{y+z}{y} p^y (1-p)^{y+z-y} \cdot \mathbb{1}_{\{0 \le y \le y+z\}} \\
&= \frac{e^{-\lambda} \cdot (\lambda \cdot p)^y \cdot (\lambda(1-p))^z}{y! \cdot z!} \mathbb{1}_{\{y \ge 0\}} \cdot \mathbb{1}_{\{z \ge 0\}}
\end{aligned}$$

(a) The p.m.f of Y can be calculated from the marginal distribution,

$$\begin{aligned}
p_Y(y) &= \sum_z p_{Y,Z}(y,z) \\
&= \sum_z \frac{e^{-\lambda} \cdot (\lambda \cdot p)^y \cdot (\lambda(1-p))^z}{y! \cdot z!} \mathbb{1}_{\{y \ge 0\}} \cdot \mathbb{1}_{\{z \ge 0\}} \\
&= \frac{e^{-\lambda} \cdot (\lambda p)^y}{y!} \sum_{z=0}^{\infty} \frac{(\lambda(1-p))^z}{z!} \mathbb{1}_{\{y \ge 0\}} \\
&= \frac{e^{-\lambda} \cdot (\lambda p)^y}{y!} \cdot e^{\lambda(1-p)} \mathbb{1}_{\{y \ge 0\}} \\
&= \frac{e^{-\lambda p} \cdot (\lambda p)^y}{y!} \mathbb{1}_{\{y \ge 0\}}
\end{aligned}$$

The p.m.f. of Z can be calculated as follows,

$$\begin{aligned}
p_Z(z) &= \sum_y p_{Y,Z}(y,z) \\
&= \sum_y \frac{e^{-\lambda} \cdot (\lambda \cdot p)^y \cdot (\lambda(1-p))^z}{y! \cdot z!} \mathbb{1}_{\{y \ge 0\}} \cdot \mathbb{1}_{\{z \ge 0\}} \\
&= \frac{e^{-\lambda} \cdot (\lambda(1-p))^z}{z!} \sum_{y=0}^{\infty} \frac{(\lambda p)^y}{y!} \mathbb{1}_{\{z \ge 0\}} \\
&= \frac{e^{-\lambda} \cdot (\lambda(1-p))^z}{z!} \cdot e^{\lambda p} \cdot \mathbb{1}_{\{z \ge 0\}} \\
&= \frac{e^{-\lambda(1-p)} \cdot (\lambda(1-p))^z}{z!} \mathbb{1}_{\{z \ge 0\}}
\end{aligned}$$

(b) To find that Y and Z are independent, consider

$$\begin{aligned}
p_Y(y) \cdot p_Z(z) &= \frac{e^{-\lambda p} \cdot (\lambda p)^y}{y!} \mathbb{1}_{\{y \ge 0\}} \cdot \frac{e^{-\lambda(1-p)} \cdot (\lambda(1-p))^z}{z!} \mathbb{1}_{\{z \ge 0\}} \\
&= \frac{e^{-\lambda} \cdot (\lambda p)^y \cdot (\lambda(1-p))^z}{y! z!} \mathbb{1}_{\{y \ge 0\}} \cdot \mathbb{1}_{\{z \ge 0\}} \\
&= p_{Y,Z}(y,z)
\end{aligned}$$

Therefore, Y and Z are independent.

3. (a) Let $X$ and $Y$ be random variables with probability densities

$$f_X(x) = \frac{1}{4}xe^{-x/2} \text{ and } f_Y(y) = \frac{1}{4}ye^{-y/2}$$

. Determine the probabilty density of $Z = X + Y$.

(b) Let $X$ and $Y$ be independent random variables with $Par(\alpha)$ and $Par(\beta)$ distributions. What is the probability density of $Z = XY$ for general $\alpha$ and $\beta$? Find the probability density when $\alpha = 3, \beta = 1$.

**Solution:**

(a)

$$f_Z(z) = \int_0^z \frac{xe^{-x/2}}{4} \cdot \frac{(z-x)e^{-(z-x)/2}}{4} dx = \frac{e^{-z/2}}{16} \int_0^z x(z-x)dx = \frac{e^{-z/2}}{16} \cdot \frac{z^3}{3} = \frac{z^3 e^{-z/2}}{96}.$$

(b)

$$f_X(x) = \frac{\alpha}{x^{\alpha+1}}, f_Y(y) = \frac{\beta}{x^{\beta+1}}$$

$$f_Z(z) = \int_1^z f_Y(\frac{z}{x}) f_X(x) \cdot \frac{1}{x} dx = \int_1^z \frac{\beta}{(\frac{z}{x})^{\beta+1}} \frac{\alpha}{x^{\alpha+1}} \cdot \frac{1}{x} dx = \frac{\beta}{z^{\beta+1}} \int_1^z \frac{\alpha}{x^{\alpha-\beta+1}} dx$$

$$\text{When } \alpha = \beta : f_Z(z) = \frac{\alpha\beta}{\alpha - \beta} \frac{1}{z^{\beta+1}} (1 - z^{\beta-\alpha})$$

$$\alpha \neq \beta : f_Z(z) = \frac{\alpha^2}{z^{\alpha+1}} \log_e(z)$$

$$\text{For } \alpha = 3, \beta = 1, f_Z(z) = \frac{3}{2z^2}(1 - 1/z^2).$$

4. Two people, $A$ and $B$, arrive at a bus stop at times distributed as

$$T_A \sim \mathcal{N}(3, 0.1), \quad T_B \sim \mathcal{N}(4, 0.2),$$

where the means are in hours (PM) and the variances are in hrs$^2$. Each person waits for 15 minutes before leaving. If they do not meet during this period, they leave. What is the probability that $A$ and $B$ meet? You can leave the answer in terms of $\Phi$, the standard normal CDF.

**Solution:** They meet if their arrival times differ by at most 0.25 hours, i.e.

$$|T_A - T_B| \leq 0.25.$$

Step 1: Distribution of difference.

Define

$$D = T_A - T_B.$$

Since $T_A$ and $T_B$ are independent normals,

$$D \sim \mathcal{N}(\mu_D, \sigma_D^2),$$

with

$$\mu_D = \mu_A - \mu_B = 3 - 4 = -1,$$
$$\sigma_D^2 = \sigma_A^2 + \sigma_B^2 = 0.1 + 0.2 = 0.3.$$

So,

$$D \sim \mathcal{N}(-1, 0.3).$$

Step 2: Probability of meeting.

We want

$$P(|D| \leq 0.25) = P(-0.25 \leq D \leq 0.25).$$

Standardize: if $Z \sim \mathcal{N}(0,1)$, then

$$P(-0.25 \leq D \leq 0.25) = \Phi\left(\frac{0.25 - (-1)}{\sqrt{0.3}}\right) - \Phi\left(\frac{-0.25 - (-1)}{\sqrt{0.3}}\right),$$

where $\Phi$ is the standard normal CDF.

5. Let $X, Y$ be continuous r.v.s with a *spherically symmetric* joint distribution, which means that the joint PDF is of the form $f(x,y) = g(x^2+y^2)$ for some function $g$. Let $(R, \theta)$ be the polar coordinates of $(X, Y)$, so $R^2 = X^2 + Y^2$ is the squared distance from the origin and $\theta$ is the angle (in $[0, 2\pi)$), with $X = R\cos\theta$, $Y = R\sin\theta$.

   (a) Explain intuitively why $R$ and $\theta$ are independent. Then prove this by finding the joint PDF of $(R, \theta)$.

   (b) What is the joint PDF of $(R, \theta)$ when $(X, Y)$ is Uniform in the unit disk $\{(x,y) : x^2 + y^2 \leq 1\}$?

   (c) What is the joint PDF of $(R, \theta)$ when $X$ and $Y$ are i.i.d. $\mathcal{N}(0,1)$?

**Solution:** (a) Intuitively, this makes sense since the joint PDF of $X, Y$ at a point $(x, y)$ only depends on the distance from $(x, y)$ to the origin, not on the angle, so knowing $R$ gives no information about $\theta$. The absolute Jacobian is $r$ , so

$$f_{R,\theta}(r,t) = f_{X,Y}(x,y)r = r \cdot g(r^2)$$

for all $r \geq 0$, $t \in [0, 2\pi)$. This factors as a function of $r$ times a (constant) function of $t$, so $R$ and $\theta$ are independent with $\theta \sim \text{Unif}(0, 2\pi)$.

(b) We have $f_{X,Y}(x,y) = \frac{1}{\pi}$ for $x^2 + y^2 \leq 1$, so

$$f_{R,\theta}(r,t) = \frac{r}{\pi} \quad \text{for } 0 \leq r \leq 1,\ t \in [0, 2\pi)$$

(and the PDF is 0 otherwise). This says that $R$ and $\theta$ are independent with marginal PDFs $f_R(r) = 2r$ for $0 \leq r \leq 1$ and $f_\theta(t) = \frac{1}{2\pi}$ for $0 \leq t < 2\pi$.

(c) The joint PDF of $X, Y$ is $\frac{1}{2\pi}e^{-(x^2+y^2)/2}$, so $g(r^2) = \frac{1}{2\pi}e^{-r^2/2}$ and the joint PDF of $(R, \theta)$ is

$$\frac{1}{2\pi} r e^{-r^2/2}.$$

This says that $R$ and $\theta$ are independent with marginal PDFs $f_R(r) = re^{-r^2/2}$ for $r \geq 0$ and $f_\theta(t) = \frac{1}{2\pi}$ for $0 \leq t < 2\pi$.

6. Consider a triangle with height $h$. Let $X$ be the distance from a point randomly chosen within the triangle to the base of the triangle. What is the CDF and the PDF of $X$?

**Solution:** Let $A = \frac{bh}{2}$ be the area of the given triangle, where $b$ is the length of the base. From the randomly chosen point, draw a line parallel to the base, and let $A_x$ be the area of the triangle thus formed.

The height of this triangle is $(h - x)$, and its base has length $\frac{b(h-x)}{h}$. Thus $A_x = \frac{b(h-x)^2}{2h}$.

For $x \in [0, h]$, we have

$$F_X(x) = 1 - P(X > x) = 1 - \frac{A_x}{A} = 1 - \frac{\frac{b(h-x)^2}{2h}}{\frac{bh}{2}}$$
$$= 1 - \left(\frac{h-x}{h}\right)^2$$

while for $x \notin [0, h]$, we have $F_X(x) = 0$

The PDF is obtained by differentiating the CDF. We have

$$f_X(x) = \frac{dF_X(x)}{dx}$$
$$= \begin{cases} \frac{2(h-x)}{h^2} & \text{if } 0 \le x \le h, \\ 0 & \text{otherwise.} \end{cases}$$

7. Let $X$ and $Y$ be independent exponential random variables with parameters $\lambda = 1$ and $\mu = 3$. Then, if $Z = min(X, Y)$ is a random variable

   (a) the value of $\mathbb{P}[X = Z]$ is?

   (b) is it true that the random variable $Z = min(X, Y)$ is independent of the event $X < Y$?

**Solution:**

(a)
$$\mathbb{P}[X = Z] = \mathbb{P}[X < Y] = \frac{\lambda}{\lambda + \mu}$$

(b) Yes. Consider the join probability distribution of the events $X < Y$ and $Z > z$

$$\mathbb{P}[X < Y, Z > z] = \mathbb{P}[z < X < Y]$$
$$= \frac{\lambda}{\lambda + \mu} e^{-(\lambda+\mu)z}$$
$$= \mathbb{P}[X < Y]\mathbb{P}[Z > z]$$

8. Let $U \sim \text{Unif}(0, 1)$. Using $U$, construct a r.v. $X$ whose PDF is $\lambda e^{-\lambda x}$ for $x > 0$ (and 0 otherwise), where $\lambda > 0$ is a constant, then $X$ is said to have a Exponential distribution; this distribution is of great importance in engineering, chemistry, survival analysis, and elsewhere.

**Solution:** The CDF of $X$ is given by $F(x) = \int_0^x \lambda e^{-\lambda t} dt = 1 - e^{-\lambda x}$, for $x > 0$ (and 0 otherwise). The inverse function is $F^{-1}(u) = -\lambda^{-1} \ln(1-u)$. So by Universality of the Uniform, $-\lambda^{-1} \ln(1-U)$ has CDF $F$.

9. Let $X$ and $Y$ have the joint pmf

$$p_{X,Y}(x,y) = \frac{C}{(x+y-1)(x+y)(x+y+1)}, \quad x, y \in \{1, 2, 3, \ldots\}.$$

Determine the following:

(a) The value of $C$.
(b) The (marginal) pmfs of $X$ and $Y$.
(c) The pmf of $Z = X + Y$.
(d) $\mathbb{P}(X = Y)$ (express the answer in summation form).

**Solution:**

(a) We require

$$\sum_{x=1}^{\infty} \sum_{y=1}^{\infty} p_{X,Y}(x,y) = 1.$$

Let $s = x + y$. Then, for each $s \geq 2$, there are $s - 1$ solutions $(x, y)$. Hence

$$\sum_{x,y} p_{X,Y}(x,y) = \sum_{s=2}^{\infty} (s-1) \frac{C}{(s-1)s(s+1)} = C \sum_{s=2}^{\infty} \frac{1}{s(s+1)}.$$

Since $\sum_{s=2}^{\infty} \frac{1}{s(s+1)} = \frac{1}{2}$, it follows that

$$C \cdot \tfrac{1}{2} = 1 \quad \Longrightarrow \quad C = 2.$$

(b) Fix $x \geq 1$. Then

$$p_X(x) = \sum_{y=1}^{\infty} p_{X,Y}(x,y) = \sum_{s=x+1}^{\infty} \frac{C}{(s-1)s(s+1)}.$$

With $C = 2$, partial fractions and telescoping yield

$$p_X(x) = \frac{1}{x(x+1)}, \qquad x = 1, 2, \ldots$$

By symmetry,

$$p_Y(y) = \frac{1}{y(y+1)}, \qquad y = 1, 2, \ldots$$

(c) For $s \geq 2$,

$$p_Z(s) = \sum_{x=1}^{s-1} p_{X,Y}(x, s-x) = \frac{C(s-1)}{(s-1)s(s+1)} = \frac{C}{s(s+1)}.$$

Thus with $C = 2$,

$$p_Z(s) = \frac{2}{s(s+1)}, \qquad s = 2, 3, \ldots$$

(d)

$$\mathbb{P}(X = Y) = \sum_{n=1}^{\infty} p_{X,Y}(n, n) = \sum_{n=1}^{\infty} \frac{2}{(2n-1)(2n)(2n+1)}.$$

This telescopes, giving

$$\mathbb{P}(X = Y) = \ln 4 - 1.$$

10. Let $X$ be Geometric($p$). Let $X_i = I(X > i)$ for $i \in \{1, 2, \ldots\}$, i.e. $X_i = 1$ if $X > i$, and 0 otherwise. Find the joint PMF of $(X_i, X_j)$ for $i, j \in \{1, 2, \ldots\}$. Are $X_i$ and $X_j$ independent? You can assume $i \leq j$ without loss of generality.

**Solution:** Let $X$ be a Geometric random variable with parameter $p$, where the probability mass function (PMF) is given by:

$$P(X = k) = (1-p)^{k-1}p \quad \text{for} \quad k = 1, 2, 3, \ldots$$

**Part (a):** For $i, j \in \{1, 2, \ldots\}$. We want to find the joint probability mass function (PMF) of $(X_i, X_j)$.

There are four possible outcomes for $(X_i, X_j)$:

1. $(X_i = 1, X_j = 1)$: This means $X > j$, since $X_j = 1$ implies $X > j$, and hence $X > i$ as well.

$$P(X_i = 1, X_j = 1) = P(X > j) = (1-p)^j$$

2. $(X_i = 1, X_j = 0)$: This means $i < X \leq j$, because $X_i = 1$ implies $X > i$, and $X_j = 0$ implies $X \leq j$.

$$P(X_i = 1, X_j = 0) = P(i < X \leq j) = P(X > i) - P(X > j) = (1-p)^i - (1-p)^j$$

3. $(X_i = 0, X_j = 1)$: This is impossible since $X_j = 1$ implies $X > j$, which contradicts $X_i = 0$ (which would imply $X \leq i$).

$$P(X_i = 0, X_j = 1) = 0$$

4. $(X_i = 0, X_j = 0)$: This means $X \leq i$, because both $X_i = 0$ and $X_j = 0$ imply $X \leq i$.

$$P(X_i = 0, X_j = 0) = P(X \leq i) = 1 - (1-p)^i$$

Thus, the joint PMF is:

$$P(X_i = a, X_j = b) = \begin{cases} (1-p)^j & \text{if } a = 1, b = 1 \\ (1-p)^i - (1-p)^j & \text{if } a = 1, b = 0 \\ 0 & \text{if } a = 0, b = 1 \\ 1 - (1-p)^i & \text{if } a = 0, b = 0 \end{cases}$$

**Part (b):** For $X_i$ and $X_j$ to be independent, we must have:

$$P(X_i = a, X_j = b) = P(X_i = a)P(X_j = b)$$

Check the case $a = 1$ and $b = 1$:

$$P(X_i = 1) = P(X > i) = (1-p)^i$$

$$P(X_j = 1) = P(X > j) = (1-p)^j$$
$$P(X_i = 1, X_j = 1) = P(X > j) = (1-p)^j$$

Thus:
$$P(X_i = 1, X_j = 1) = (1-p)^j \neq P(X_i = 1)P(X_j = 1) = (1-p)^i(1-p)^j$$

Therefore, $X_i$ and $X_j$ are **not independent**.

11. Consider two continuous random variables $Y$ and $Z$, and a random variable $X$ that is equal to $Y$ with probability $p$ and to $Z$ with probability $1-p$. (a) Show that the PDF of $X$ is given by
$$f_X(x) = pf_Y(x) + (1-p)f_Z(x)$$
(b) Calculate the CDF of the two-sided exponential random variable that has PDF given by
$$f_X(x) = \begin{cases} p\lambda e^{\lambda x}, & \text{if } x < 0 \\ (1-p)\lambda e^{-\lambda x}, & \text{if } x \geq 0 \end{cases}$$
where $\lambda > 0$ and $0 < p < 1$.

**Solution:** (a) By the total probability theorem, we have
$$F_X(x) = \mathbf{P}(X \leq x) = p\mathbf{P}(Y \leq x) + (1-p)\mathbf{P}(Z \leq x) = pF_Y(x) + (1-p)F_Z(x)$$
By differentiating, we obtain
$$f_X(x) = pf_Y(x) + (1-p)f_Z(x)$$
(b) Consider the random variable $Y$ that has PDF
$$f_Y(y) = \begin{cases} \lambda e^{\lambda y}, & \text{if } y < 0 \\ 0, & \text{otherwise} \end{cases}$$
and the random variable $Z$ that has PDF
$$f_Z(z) = \begin{cases} \lambda e^{-\lambda z}, & \text{if } y \geq 0 \\ 0, & \text{otherwise} \end{cases}$$
We note that the random variables $-Y$ and $Z$ are exponential. Using the CDF of the exponential random variable, we see that the CDFs of $Y$ and $Z$ are given by
$$F_Y(y) = \begin{cases} e^{\lambda y}, & \text{if } y < 0 \\ 1, & \text{if } y \geq 0 \end{cases}$$
$$F_Z(z) = \begin{cases} 0, & \text{if } z < 0 \\ 1 - e^{-\lambda z}, & \text{if } z \geq 0 \end{cases}$$
We have $f_X(x) = pf_Y(x) + (1-p)f_Z(x)$, and consequently $F_X(x) = pF_Y(x) + (1-\text{ p})\ F_Z(x)$. It follows that
$$F_X(x) = \begin{cases} pe^{\lambda x}, & \text{if } x < 0 \\ p + (1-p)\left(1 - e^{-\lambda x}\right), & \text{if } x \geq 0 \end{cases}$$
$$= \begin{cases} pe^{\lambda x}, & \text{if } x < 0 \\ 1 - (1-p)e^{-\lambda x}, & \text{if } x \geq 0 \end{cases}$$